CMPE-255 Data Mining Project Report DRAFT

Title - Mental Health & Crisis Signal Mining on Social Media

GitHub Link: https://github.com/NateyLB/CMPE255Project

1. Project Team

Course	CMPE 255 – Section 34, Spring 2026
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2. Dataset

Three publicly available, CC0-licensed datasets from Kaggle are used. No IRB or data-use agreement is required; all datasets are loaded directly via the Kaggle API using pandas.

Dataset	Size	Labels	Role in Project
Reddit Mental Health Dataset	~50,000 posts	Depression, Anxiety, Suicidal Ideation, Control	Primary training/test set
Dreaddit (Stress Analysis)	3,553 posts	Abuse, Anxiety, Financial Stress, PTSD, Social	Supplementary stress signal
Depression & Anxiety on Reddit	~37,000 posts	Depression, Anxiety, Control	Held-out generalization test

Dataset URLs:

- Reddit Mental Health Dataset: https://www.kaggle.com/datasets/neelghoshal/reddit-mental-health-data
- Dreaddit: https://www.kaggle.com/datasets/ruchi798/stress-analysis-in-social-media
- Depression & Anxiety on Reddit: https://www.kaggle.com/datasets/infamouscoder/mental-health-social-media

3. Abstract - Project Idea

Mental health conditions such as depression, anxiety, and suicidal ideation affect hundreds of millions of people worldwide, yet remain significantly underdiagnosed due to stigma and limited access to care. Social media platforms, particularly Reddit, have emerged as spaces where individuals can openly express their struggles, creating a lot of text that can be used for automated analysis. This project investigates whether data mining and natural language processing techniques can reliably detect and classify mental health crises from Reddit posts. We will build a multi-class classification pipeline that classifies posts as depression, anxiety, suicidal ideation, or a neutral/control. Classical machine learning approaches (Logistic Regression, Linear SVM, XGBoost with TF-IDF features) will be systematically compared against a fine-tuned DistilBERT transformer to show the benefit of deep contextual representations. Latent Dirichlet Allocation (LDA) topic modeling will be applied to surface interpretable latent themes, and SHAP values will be used to explain individual predictions at the token level. A lightweight Streamlit web application will provide a live interface. The system will be trained and evaluated on three freely available Kaggle datasets totaling approximately 90,000 annotated Reddit posts.

4. Introduction

4.1 Motivation

The global burden of mental illness is staggering. The World Health Organization estimates that one in eight people worldwide lives with a mental health disorder, yet the majority never receive a diagnosis or treatment. Stigma, cost, and geographic barriers to care mean that many individuals only seek help after a crisis has already occurred.

Social media platforms, particularly Reddit, have become an important informal support channel. Users post candidly about depressive episodes, anxiety, and suicidal ideation in dedicated subreddits (r/depression, r/anxiety, r/SuicideWatch). This creates a large, naturally labeled set of mental health posts that can be used for early identification of at-risk individuals.

4.2 Objective

This project has three core objectives:

- Build a robust multi-class NLP classification pipeline that properly classifies depression, anxiety, suicidal ideation, and neutral posts with a high macro-averaged F1 score.
- Compare classical bag-of-words models (TF-IDF + Logistic Regression, SVM, XGBoost) against a contextual transformer model (fine-tuned DistilBERT), providing empirical analysis of the precision-recall trade off and computational cost.
- Ensure model interpretability through SHAP token-level attributions and LDA topic visualization, making results trustworthy and auditable for potential clinical use.

4.3 Technical Challenges

Several challenges shape the design of this pipeline:

- Class Imbalance: Suicidal ideation posts are significantly rarer than depression or neutral posts. We will address this with SMOTE oversampling and class-weighted training, evaluating both approaches.
- Lexical Overlap: Language used in depression and anxiety posts overlaps considerably, making fine-grained multi-class separation very important.
- Distributional Shift: Models trained on one subreddit community may not generalize to another. We will use the third dataset (Depression & Anxiety on Reddit) as a held-out generalization test.
- Interpretability vs. Accuracy Trade-off: DistilBERT achieves much higher classification accuracy but is harder to explain; SHAP bridges this gap.

5. Problem statement

Given a Reddit post text P, the goal is to assign it to one of K = 4 classes:

- C_1 = Depression
- C_2 = Anxiety
- C_3 = Suicidal Ideation
- C_4 = Neutral / Control

Formally, we seek a classifier $f: X \rightarrow \{C_1, C_2, C_3, C_4\}$ where X is the space of raw post texts, trained to maximize macro-averaged F1 score over a stratified 80/20 train-test split. We will deliberately use macro-averaging (rather than accuracy or micro-F1) because the classes are unequal, and we want the model to perform well across all classes, not just the majority.

Secondary goals include: discovering the latent topical structure of each class via LDA, and explaining individual classifications via SHAP token attributions.

6. Description of Steps

Step 1 – Data Acquisition & Cleaning

All three datasets are downloaded using the Kaggle API and loaded with pandas. Cleaning operations include: removal of duplicate posts, stripping of HTML tags, URLs, and special characters, lowercasing, and mapping of label strings to integer class indices. Class imbalance is handled by comparing SMOTE oversampling against class-weighted training loss; a stratified 80/20 split ensures label proportions are preserved in both train and test sets.

Step 2 – Exploratory Data Analysis (EDA)

EDA characterizes the dataset before modeling. Key analyses include: label distribution bar charts, post-length distributions (word count per class), vocabulary size per class, top-n unigram/bigram frequency plots, and class-stratified word clouds. These visualizations reveal lexical fingerprints for each condition and motivate subsequent feature engineering decisions.

Step 3 – Feature Engineering

Four types of features are constructed:

- TF-IDF Vectorization: Unigrams and bigrams, with max_features = 50,000 and sublinear TF scaling.
- LIWC-Inspired Lexicon Features: Counts of words from emotional, cognitive, and social process categories to capture patterns.
- Post-Level Statistics: Word count, sentence count, punctuation density, capitalization ratio. These will be used as surface signals that can be correlated with emotional intensity.
- LDA Topic Probabilities: A K-topic LDA model (k selected via coherence scores) generates per-post topic probability vectors used as supplementary features.

Step 4 – Model Training & Comparison

Four models are trained on identical train splits and evaluated on the same held-out test set:

- Logistic Regression (C = 1.0, L2 regularization) with TF-IDF features, this will be our baseline.
- Linear SVM (kernel = linear, C tuned via 5-fold cross-validation) with TF-IDF features.
- XGBoost with TF-IDF, and engineered feature concatenation, tuned via Bayesian search.
- DistilBERT (distilbert-base-uncased) fine-tuned for 3 epochs with AdamW optimizer (lr = 2e-5, batch size = 32) using HuggingFace Transformers.

Step 5 – Topic Modeling

LDA is applied with k ranging from 10 to 15 topics. The optimal k will be selected by maximizing the C_v coherence score. Topic-class associations will be visualized using pyLDAvis to reveal which latent themes (e.g., isolation, hopelessness, physical symptoms) co-occur with which diagnostic labels.

Step 6 – Evaluation & Explainability

Each model will be evaluated using macro-averaged F1, per-class precision/recall, ROC-AUC (one-vs-rest), and confusion matrices. SHAP TreeExplainer (for XGBoost) and SHAP partition explainer (for DistilBERT) will be used to generate token-level importance visualizations. A lightweight Streamlit demo will enable live text input and risk category prediction with a confidence score.

7. Related work

7.1 Classical NLP Approaches to Mental Health Detection

Coppersmith et al. [1] established an influential baseline using Twitter data in the CLPsych 2015 shared task, demonstrating that TF-IDF-weighted n-gram features combined with standard classifiers (SVM, Logistic Regression) can reliably distinguish depression and PTSD posts from controls. This work motivates our baseline experimental setup.

7.2 Stress and Multi-Domain Signal Analysis

Turcan and McKeown [2] introduced Dreaddit, a Reddit dataset spanning five stress domains (abuse, financial, PTSD, social, anxiety). Their findings show that stress detection benefits from domain-aware modeling and that cross-domain generalization is significant. We incorporate Dreaddit as a complementary stress signal to enrich our multi-label analysis.

7.3 Deep Learning and Transformer Models

Ji et al. [3] conducted a comprehensive survey of machine learning methods for suicidal ideation detection, showing that BERT-based fine-tuned models consistently outperform classical ML baselines on mental health text classification tasks (macro-F1 gains of 5–15 percentage points). This validated our plan for the inclusion of DistilBERT as the deep learning comparator.

7.4 Model Interpretability

A growing body of work emphasizes interpretability as a prerequisite for clinical deployment. SHAP (Shapley Additive Explanations) has been applied to mental health NLP models to surface the lexical drivers of predictions [3], enabling human auditing of model decisions. We will adopt SHAP as our primary interpretability tool because of this.

8. Solution

Our solution is an end-to-end NLP pipeline with six stages:

- Data Integration & Cleaning: Unified preprocessing pipeline merging three Kaggle Reddit datasets into a single multi-class corpus with standardized labels.
- Feature Representation: Dual-track features: TF-IDF bag-of-words for classical models, and raw tokenized sequences for DistilBERT fine-tuning.
- Multi-Model Comparison: Four models trained under identical conditions, enabling controlled removal: LR (baseline) → SVM → XGBoost → DistilBERT (transformer).
- Topic Modeling: LDA with coherence-based k selection, providing class-level interpretable topic clusters as a supplementary analytical layer.
- Explainability: SHAP token-level attributions for the best performing model.

- Deployment: Streamlit demo application for live inference.

The pipeline is implemented in Python using scikit-learn, XGBoost, HuggingFace Transformers, gensim (LDA), shap, and streamlit. Code will be in a shared GitHub repository.

9. List of experiments

Experiment	Model / Method	Feature Input	Expected Macro-F1
EXP-1 (Baseline)	Logistic Regression	TF-IDF (unigrams + bigrams)	0.72 – 0.78
EXP-2	Linear SVM	TF-IDF (unigrams + bigrams)	0.75 – 0.80
EXP-3	XGBoost	TF-IDF + engineered features	0.79 – 0.83
EXP-4	DistilBERT (fine-tuned)	Raw token sequences	0.85 – 0.90

All experiments use:

- Train/Test Split: Stratified 80/20 split.
- Cross-Validation: 5-fold stratified CV on the training set for hyperparameter tuning.
- Class Imbalance: SMOTE for classical models; class_weight="balanced" evaluated as an alternative.
- Feature Removal: TF-IDF only vs. TF-IDF + engineered features (EXP-3) to isolate the contribution of hand-crafted features.

10. Proposed Evaluation Metrics and Justification

Metric	Justification
Macro-averaged F1	Treats each class equally regardless of support; important for imbalanced multi-class distributions (depression, anxiety, suicidal ideation, neutral).
Per-class Precision / Recall	Precision controls false alarms; recall ensures at-risk cases are not missed. Both are clinically significant.

ROC-AUC (per class, OvR)	Measures separability between each class and the rest; useful for ranking models beyond a single threshold.
Confusion Matrix	Provides a full error breakdown, which classes are overlapped with which, enabling targeted preprocessing or model refinement.
LDA Coherence Score (C_v)	Determines the optimal number of latent topics k by measuring the semantic coherence of top words per topic.
SHAP Token Importance	Provides model interpretability by matching predictions to individual tokens; essential for clinical trustworthiness.

Primary ranking metric: Macro-averaged F1. Secondary: ROC-AUC. Models will be compared on both metrics, with the best macro-F1 model selected as the target for SHAP analysis.

11. Results

11.1 Exploratory Data Analysis – Initial Findings

Three datasets were loaded and explored. The full EDA notebook is available in the GitHub repository at notebooks/01_EDA.ipynb.

Dataset Sizes (after dropping null text rows):

- Reddit Mental Health: 5,957 posts across 5 target classes, roughly balanced (~1,200 posts each)
- Dreaddit: 3,553 posts (2,838 train + 715 test) across 10 subreddits, binary stress labels (1,857 stressed / 1,696 not stressed), includes 110 pre-computed LIWC linguistic features
- Depression & Anxiety on Social Media: 20,000 Twitter posts, perfectly balanced binary labels (10,000 each)

Key Observations:

- The Reddit dataset is well-balanced across all 5 classes, reducing the need for SMOTE oversampling on the primary task.
- Post lengths differ significantly across datasets: Reddit posts average hundreds of words while Twitter posts are substantially shorter due to platform character limits.
- Top unigram and bigram analysis reveals lexical overlap between classes (e.g., "feel," "people," "time" appear across categories), confirming the classification challenge noted in Section 4.3.
- The Dreaddit dataset includes pre-computed LIWC features covering emotional, cognitive, and social process categories, which can supplement our feature engineering.

- Subreddit-level stress rates in Dreaddit vary considerably, with PTSD and domestic violence subreddits showing the highest proportion of stressed posts.

Note: Dataset specifications and class label mappings are being refined based on these EDA findings and will be updated in Check-in 2.

11.2 Baseline Model (Placeholder)

Note: Full results, corrected dataset specifications, and baseline model outputs will be reported in Check-in 2.

12. Conclusions

13. Limitations

- Dataset Source Homogeneity: All data originates from Reddit, limiting generalizability to other social platforms (Twitter, Facebook) or clinical interview transcripts.
- Self-Report Bias: Reddit posts are self-reported and unverified by clinical professionals. Labels are derived from subreddit membership, not clinical diagnosis, introducing label noise.
- Temporal Generalization: Mental health communication on social media changes over time (e.g., post-COVID language shifts). Models trained on historical data may not perform well on future posts.
- GPU Access Constraints: DistilBERT fine-tuning requires a GPU; limited cloud GPU credits may constrain the number of hyperparameter search iterations.
- Class Overlap: Depression and suicidal ideation share significant overlap, and there is a very fine line between these two, potentially exacerbating confusion between adjacent classes.
- Ethical Scope: This project does not deploy predictions in a real clinical context; the Streamlit demo is for demonstration purposes only and should not be used for clinical screening.

14. Future work

- Cross-Platform Generalization: Extend the dataset to include Twitter and Facebook mental health posts to test multi-platform generalization.
- Longitudinal Analysis: Model temporal trajectories of mental health signals for individual users (post sequence modeling) using RNNs or temporal transformers.
- Multimodal Signals: Incorporate user engagement metadata (post frequency, upvote ratio, reply depth) as additional behavioral features.
- Clinical Validation: Collaborate with mental health professionals to validate LDA topic clusters and SHAP-identified keywords against established symptoms (DSM-5).
- Active Learning: Implement a feedback loop where flagged predictions are reviewed and corrected to improve model accuracy over time, which can reduce labeling cost.
- Larger Transformer Models: Evaluate MentalBERT or mental-RoBERTa pre-trained on mental health data, which may offer further gains over general-domain DistilBERT.

15. Contributions

Team Member	Role	Key Responsibilities
Jesigga	**Data Engineer**	Data acquisition (Kaggle API), preprocessing pipeline (cleaning, tokenization, SMOTE for class balancing), EDA (class distribution plots, word length analysis, label statistics)
Ravikumar	**NLP Feature Engineer**	TF-IDF vectorization, n-gram analysis, LDA topic modeling, pyLDAvis visualization, word cloud generation per class, LIWC-inspired lexicon features
Gunanidhi:	**ML Modeling Lead**	Train and tune Logistic Regression, SVM, and XGBoost classifiers; implement 5-fold cross-validation; generate confusion matrices and per-class F1/precision/recall comparisons
Nathan	**Deep Learning & Deployment**	Fine-tune DistilBERT (HuggingFace), generate SHAP explainability plots, build Streamlit demo app, lead paper write-up, and slide deck preparation

16. References

[1] Coppersmith, G., Dredze, M., Harman, C., Hollingshead, K., & Mitchell, M. (2015). CLPsych 2015 shared task: Depression and PTSD on Twitter. Proceedings of the 2nd Workshop

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[2] Turcan, E., & McKeown, K. (2019). Dreaddit: A Reddit dataset for stress analysis in social media. Proceedings of the 9th International Workshop on Health Text Mining and Information Analysis (LOUHI), pp. 97–107. https://aclanthology.org/D19-6213/

[3] Ji, S., Pan, S., Li, X., Cambria, E., Long, G., & Huang, Z. (2021). Suicidal ideation detection: A review of machine learning methods and applications. IEEE Transactions on Computational Social Systems, 8(1), 214–226. https://doi.org/10.1109/TCSS.2020.3021467